

CLASSES OF COMPUTERS

Computers can be classified based on the following parameters:

- ◆ Type of data processed
- ◆ The size
- ◆ The Purpose or scope
- ◆ The age of technology

Classification by type of data processed

- 1 Digital computers
- 1 Analog computers
- 1 Hybrid computers

1. Analog Computers

Analog is the Greek word, which means similar. So, in analog computers, the similarities between any two quantities are measure by electrical voltages or current.

The analog computers operate by measuring instead of counting. The analog computer works on the supply of continuous electrical signals. The display is also continuous. Its output is in the form of graphs.

Analog Signal: An analog signal is a continuous variable electromagnetic wave. It can consume an infinite number of voltage or current values.

Characteristics of analog computer

The analog computer has the following important characteristics:

- a) It operates by measuring.
- b) It requires physical analog.
- c) It functions on continuously varying quantities.
- d) The output is usually represented in the form of graph.
- e) In analog, the calculations are first converted in equation and later converted into electrical signals.
- f) The accuracy of the output is poor.
- g) It has limited memory space.
- h) It is not versatile, i.e., it has limited application.
- i) The speed of analog computer is low.
- j) Not suitable for business and industry.

2. Digital Computers

These computers work with quantities represented as digits. They operate on discrete quantities. In digital computer, both numeric and non-numeric information are represented as strings of digits. These computers use binary codes, 0's and 1's, to represent the information. The information is given to the computer in the form of discrete electrical signals. The basic operation performed by a digital computer is addition. Hence, the other operations such as multiplication, division, subtraction and exponentiation are first converted into "addition" and then computed.

Characteristics of digital computer

The digital computer has the following characteristics:

- a) It operates by counting.
- b) It functions on discrete numbers.
- c) The calculations are converted into binary numbers i.e., 1s and 0s.
- d) The output is represented in the form of discrete values.
- e) Its accuracy is good
- f) It has large memory space
- g) It is versatile in nature and is suitable for a number of applications
- h) In digital computer, the numbers, words and symbols can be used.
- i) Its processing speed is high.
- j) It is highly suitable for business application.

Analog VS. Digital computer

S/NO	Analog computer	Digital computer
01	It operates by measuring	It operates by counting
02	It requires physical analog	It functions on discrete numbers
03	The calculations are first converted to equations and later converted into electrical signals	The calculations are converted in to binary numbers(1s and 0s)
05	Less accurate	More accurate
06	Less speed	High speed
07	Limited memory is available	More memory is available
08	Lacks flexibility	Highly flexible
09	Cannot process alpha-numeric information	Can process alpha number information
10	Limited application	More number of applications
11	Not suitable for business applications	Highly suitable for business applications

3. Hybrid computers:

A hybrid computing system is a combination of desirable features of analog and digital computers. Mostly used for automatic operations of complicated physical processes and machines. Nowadays analog-to-digital and digital-to-analog converters are used for transforming the data into a suitable form for either type of computation.

For example, in the hospital's automated intensive care unit, analog devices might measure the patient's temperature, blood pressure, and other vital signs. These measurements which are in analog might then be converted into numbers and supplied to digital components in

the system. These components are used to monitor the patient's vital signs and send signals if any abnormal readings are detected. Hybrid computers are mainly used for specialized tasks.

Classification by Purpose or Scope

Using scope computers can be classified into two broad categories:

(a) *Dedicated or Special purpose computers*: These are computers that are designed to carry out only specified task. The series of instructions that these types of computer follow to carry out its operation is in-built and cannot be modified. E.g. word processor, robots used in car manufacturing plants.

(b) *General Purpose Computers*: These are computers designed to perform a wide variety of operations. They can be programmed to carry out scientific oriented applications or business oriented ones just by changing the series of instructions in its memory.

Classification by Size

Using physical size as a factor, the following types of computers can be identified:

(a) *Micro-Computers*: These are computers that are small in size which can be placed on the desk or lap or palm.

(b) *Mini Computers*: These are large computers that support multi users. Their speed of operations is high compared to micros.

(c) *Mainframe Computers*: These are very large computers that support multi users. Their speed of operations and memory capacity is larger than that of a mini computer.

Classification by Age of Technology

This classification is also referred to as generations of computers. See our discussion on generations of computers in the above discussion.

AN OVERVIEW OF COMPUTER ORGANIZATION

Computers are often compared to human beings since both can accept data, store, work with it, retrieve and provide information. The main difference is that human beings can perform all of these actions independently. While a computer requires a program to perform an assigned task. A Computer system is a set of physical equipment (hardware) and appropriate programs (software) logically connected and working together to meet the desired need. The hardware components include input

devices, processors, storage devices, and output devices. The software items are programs and operating aids (systems) so that the computer can process data.

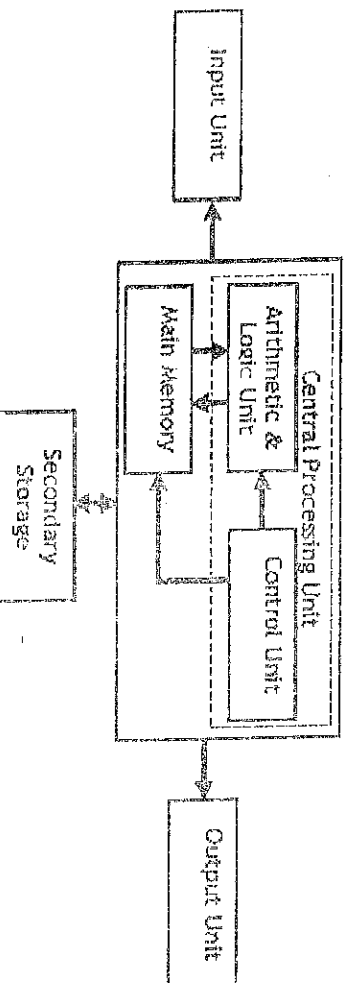


Fig. 39: Components of a computer system

In a nutshell, if you use a desktop computer as an example you might already know that here is no single part called the “computer.” A computer is a system of many parts working together. The physical parts, which you can see and touch, are collectively called hardware. Software, on the other hand, refers to the instructions, or programs.

Functional Units of a Computer System

A computer system consists of five major categories of hardware components devices.

1. Processing devices ii) Primary memory iii) Secondary storage devices iv) Input devices
2. Output devices

Input devices: A computer uses input devices to accept the data and program. Input devices allow communication between the user and the computer e.g. keyboard, mouse, light pen, touch screen, etc.,

Output devices: These devices take machine coded output results from the processor and convert them into a form that can be used by human beings e.g. monitors and printers.

Processing devices: CPU is the brain of any computer system that takes all major decisions, and all sorts of calculations. It consists of arithmetic and logic units, control unit, and internal memory (registers). *The control unit of the CPU coordinates the action of the entire system. The Control Unit controls all the hardware operations. The arithmetic and logic units performing addition, subtraction, division, and multiplication as well as some logical operations.* The instructions and data are stored in the main memory (RAM) so that the processor can directly fetch and execute them. While between any two

components of the computer system, there is a pathway called a bus which allows for the data transfer between them.

Memory devices: The computer stores the program and data that are currently being used in the main memory (RAM). The main memory holds data and programs only temporarily. Secondary storage devices can hold more storage than the main memory and are much less expensive.

ROLES OF COMPUTERS IN A SOCIETY:

In this digital age of information technology, it has become a necessity for everyone to know about computers. This is based on the fact that the problems that we must solve are no longer solvable without the aid of computers. The application of computers in the society has led to many changes in education, government, health, homes, industry, law, medicine, scientific research, social service, and even arts like music, movies, and paintings.

Computers at home: Many homes have multiple computers used for Internet communication, shopping, business, entertainment, schoolwork, and finances.

Business: A computer has a high speed of calculation, diligence, accuracy, reliability, or versatility which has made it an integrated part in all business organizations for; payroll calculations, budgeting, sales analysis, financial forecasting, managing employee database maintenance of stocks, etc.

Banking: Today, banking is almost totally dependent on computers. Online accounting facilities, including checking current balance, making deposits and overdrafts, checking interest charges, shares, and trustee records are computer-based. ATMs that are completely automated are making it even easier for customers to deal with banks.

Insurance: Insurance companies, finance houses, and stockbroking firms are widely using computers for maintaining a database of all clients with information showing procedure to continue with policies, starting date of the policies, next due instalment of a policy, maturity date, interests due, survival benefits and bonus.

Education: Computer literacy is required at all levels in providing a lot of facilities in the education system. The computer provides a tool in the education system known as CBI (Computer Based Education) which involves control, delivery, and evaluation of learning. It is used to prepare a database about the performance of a student and analysis is carried out on this basis.

Marketing: Computers are used in marketing for advertising and selling more products.

Healthcare: Computers have become an important part of hospitals, labs, and dispensaries, to keep the record of patients and medicines and to diagnose different diseases. Such as:

- ▢ Diagnostic System to collect data and identify the cause of illness.
- ▢ Patient Monitoring System to check the patient's signs for abnormality such as in Cardiac Arrest,
- ▢ Pharma Information System to check drug labels, expiry dates, harmful side effects, etc.
- ▢ Surgery is used in performing surgery.

Engineering Design: Computers are widely used for engineering purposes.

Such as:

- ▢ Structural Engineering for the design of ships, buildings, budgets, airplanes, etc.
- ▢ Industrial Engineering for design, implementation, and improvement of integrated systems of people, materials, and equipment.
- ▢ Architectural Engineering for planning towns, designing buildings, determining a range of buildings on a site using both 2D and 3D drawings.

Military: The military employs computerized control systems in defines. Modern tanks, Missile Control. Military Communication, Military Operation and Planning, and Smart Weapons are computer-based.

Communication: Communication is a way to convey a message, an idea, a picture, or speech by the sender to the person for whom it is meant. A computer helps in transmitting the message via E-mail, Chatting, Usenet, FTP, Telnet teleconferencing (conference call) and Video-conferencing

Government: Governments are the first computer users as they play an important role in government services. Such as Budget planning, Sales tax calculation, Income tax department, track data for population, Computerization of voters lists, and Weather forecasting. Police officers, dissemination of policies, etc.

DATA COMMUNICATION AND NETWORK

Communication means an exchange of information between two or more parties and it can be exchange in a variety of way, that might be in the form of word, letters, messages drawing, body movement or any other symbols that represents the idea which make to other understandable.

Data communication entails electronically exchanging data or information. It is the movement of computer information from one point to another by means of electrical or optical transmission system. This system often is called data communication networks.

Data Communication systems can therefore be defined as the combinations of facilities including software to aid data movement from one location to the other electronically;

In today's computing world, data refers to facts, statistics, pictures, voice and other information that is digitally coded and intelligible to variety of electronic machines.

The key technology of the information age is computer communications. The value of high-speed data communication network is that it brings the message sender and receiver closer together in time. Data communication and networking is a truly global area of study. It facilitates more efficient use of computers and improves the day to day control of business by providing faster information flow. The information society where information and intelligence are the key drivers of personal, business, and national success has arrived. Data communication is the principle enabler of the rapid information exchange and will become more important than the use of computer themselves in the future. To be more familiar that what is data communication and its capabilities along with its limitation, we have to be familiar with the data communication media, components of data communication

Communication Media

It is necessary to have a pathway or the media to transmit data from one point to another. The term media means the device that transmits the voice or the data form one point to another. Many different types of transmission media are in use today for example, copper wire, glass or plastic cables, which are called fiber optic cables or air transmission (radio, infrared, microwave, or satellite). There are two types of media; one is *guided media* and another is *radiated*.

In *Guided media* the messages flow through the physical media like twisted pair-wire, coaxial cable, fiber optic cables, the media guided is the signal.

In *Radiated media* messages is broadcast through the air such as infrared, microwave, or satellite.

Data Transmission and Communication Device

Data is the specific individual facts or the facts, from which decision can be made; it is the raw facts that can be processed into accurate and relevant information. *Digital* and *analog* is the two types of fundamental data;

Digital and *analog* transmission have the following features;

- ❖ Computers produce the digital data whereas telephone produces analog data
- ❖ Digital data are binary either on or off and the analog data is the sound wave
- ❖ Analog signals are continuous waves that carry information by the characteristics of the waves. A analog transmission is the transmission of a continuously variable signal as opposed to a direct on/off signal.
- ❖ The traditional way of transmits telephone signal is an example of analog transmission. Digital signal is a *discontinuous signal* whose various states are discrete intervals apart, such as +15 volts and -15 volts. Most computers transmit the data in a digital form over the digital circuit to the attached device for example to the "printer" same as analog voice data also can transmit to the analog form through the telephone networks.

Data can be transmitted from one specific form to another. For example; as we know most of computer produces the digital data; we can convert digital data to analog voice data by using the special device called **MODEM** through the analog telephone circuit.

Data transmission may be either *asynchronous* or *synchronous*.

Asynchronous transmit only one character at a time and during transmission the character is preceded by a start bit and followed stop bit that lets the receiving device know where a character begins and ends. Asynchronous transmission is inherently inefficient due to the additional overhead required for start and stop bits, and the idle time between transmissions. It is therefore generally used for only low speed data transmission.

Synchronous transmission is a form of transmission in which data is sent as a block

using frames or packets. In Synchronous transmission a group of character is sent to communication link in a continuous bit stream while data transfer is controlled by a timing signal initiated by the sending device. Used for high speed transmission.

ON LINE DATA TRANSMISSION

This means that communication lines are directly connected to the computer either by means of a multiplexor or (Transmission controller) or by means of interface unit. The interface unit scans the communication lines frequently to dictate those that are ready to send or receive data. When a line is ready to transfer data, the scanning ceases and the channel number of the line is signaled to the CPU if the processor is in a position to accept data, then transmission begins.

OFF-LINE DATA COMMUNICATION

This type of data transmission indicates that the communication lines are not Connected directly to the computer. Key-to-disk are often used for off-line preparation and transmission of data in remote job-entry-system.

ELEMENTS OF DATA COMMUNICATION

- (a) Information source
- (b) Message
- (c) Transmitter
- (d) Signal
- (e) Receiver
- (f) Information Destination

Information Source: Originator of messages wishing to convey information to another person.

Message: This is the idea/information that the sender wants to pass across to the receiver at a particular destination.

Transmitter: Wireless operator transmitting messages by means of a mouse key connected to transmitting scl.

Signal: The signal produced by the transmitting scl when the mouse key is depressed.

Receiver: Wireless operator receiving message by means of a receiving scl.

Information Destination: Receipt of, message by addressee.

EQUIPMENT/COMPONENTS REQUIRED FOR DATA TRANSMISSION

(a) Multiplexor (b) Front-End-Processor (c) Concentrator (d) Modem

Modems

MODem Stands for Modulator Demodulator is used to convert analogue signal to digital and vice versa. A modem is a device that modulates and demodulates the signals. Modulation means the conversions of data from digital signal to analog signal and demodulates means the conversion of data signal from analog to digital signal. Modems are always use in pairs. The unit at the sending end converts the digital signal to analog signals for transmission through the analog circuit; at the receiving end, another modem converts analog to digital for the receiving computer. Modem transmission rate is measured in bits-per-second and it usually expresses bps. The amount of data actually transferred for one system to another is depends upon the speed of the modem.

Front-End Processor (FEP): A front-end processor is the special purpose device; this specialized computer manages all routing communications with peripheral devices. For large mainframe installations, a front-end processor is typically a special purpose minicomputer. The primary application of the FEP is to serve as the interface between the host/client computer and the data communication network with its thousands of terminals or microcomputers. It can handle coding and decoding data, error detection, retrieval of messages, interpreting, and processing of the control information, addressing, routing message to the right computer, and breaking the long message in to small packets. Front-end processor can provide up to 30 percent additional processing time for mainframe.

Multiplexors

Multiplexors is an electronic device that allows a single communications channel to carry data transmissions simultaneously from many sources. Multiplexing a data communication means that two or more information is sent simultaneously via the communication circuit. Multiplexors is transparent, when the circuits are multiplexed at one end and DDM-multiplexed at the other, each user's terminal or microcomputer thinks it has its own separate connection to the host computer. The objective of multiplexor is to reduce communication costs by allowing the more efficient use of circuits through sharing.

TYPES OF TRANSMISSION

Simplex Channel: In this type of communication, transmission takes place in one direction. The system known as Teletex, the sending of data alongside a television signal, is a good example of a simplex channel. Here the recipient has no formal means of communicating with the originator to ask for a repeat if there is an error.

Half-Duplex: This communications is bi-directional, but transmission moves in one direction at a time. The example is in radio telephone conversation. A protocol using the word "over" to allow the direction of data flow to change is used.

Full-Duplex: This allows bi-directional communication simultaneously. This is used for normal telephone conversation where both the initiator and the receiver can talk at the same time.